

Lecture 20: Self-gravitation and rotation

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Michaelmas Term, 2026

Outline and motivation

In previous lectures we have discussed how ~~that~~ travel time and waveform observations can be used to determine variations in elastic wave speeds within the Earth. Nothing has, however, been said about **density**, with this parameter being key to understanding both the Earth's interior composition and dynamics. To learn about density, we need to study the ~~long-period free oscillations~~ long-period free oscillations of the Earth. This is because **self-gravitation**, and hence density, plays a vital role in their physics. Within this lecture we will return to basic elastodynamics, and show how gravitational forces can be included into the equations of motion. We also explain how the **Earth's rotation** enters into the problem. Having obtained the equations of motion, we perform a **scaling analysis** to determine the relative importance of the different terms, and conclude that at short enough periods the effects of self-gravitation and rotation can be safely neglected, and hence the earlier parts of this course remain valid.

A review of Newtonian gravitation

Consider the motion φ of an elastic body relative to a reference body M , and write M_t for the instantaneous volume of space occupied at time t . At this instant in time, the gravitational potential of the body satisfies Poisson~~equation~~

$$\nabla^2 \phi = 4\pi G \varrho.$$

's equation

$$\nabla^2 \phi = 4\pi G \varrho, \tag{1}$$

in $\mathbb{R}^3$, where G is the universal gravitational constant, and ϱ is the density. The gravitational acceleration associated with this potential is given by

$$g_i = - \frac{\partial \phi}{\partial x_i}$$
$$\nabla^2 \phi = 4\pi G \varrho. \tag{2}$$

~~The above Poisson~~Poisson's equation is subject to the conditions that ϕ tends to zero at infinity, while across any jumps in density we have the continuity conditions

$$\begin{aligned} [\phi]_{-}^{+} &= 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0 \\ [\phi]_{-}^{+} &= 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0, \end{aligned} \quad (3)$$

where $[\cdot]_{-}^{+}$ denotes the jump in a quantity in the direction of ~~the~~ outward normal. The closed form solution to this problem is given by

$$\begin{aligned} \phi(\mathbf{y}, t) &= -G \int_{M_t} \frac{\rho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \\ \phi(\mathbf{y}, t) &= -G \int_{M_t} \frac{\rho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \end{aligned} \quad (4)$$

and differentiating, we ~~similar find~~

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\rho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'.$$

~~similarly find~~

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\rho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'. \quad (5)$$

For the Earth, the equilibrium gravitational acceleration and the pressure that it produces are shown in Fig. 1. These were calculated from the radial density profile of the model PREM; where that density comes from is a question we return to in Lecture 24, since it is the free oscillations discussed there that provide the main constraint on it. Note that g is close to 10 m s^{-2} throughout the mantle, reaching its largest value at the core-mantle boundary, and decreases roughly linearly through the core.

It will be convenient to define the **referential gravitational potential** by

$$\begin{aligned} \zeta(\mathbf{x}, t) &= \phi[\varphi(\mathbf{x}, t), t], \\ \zeta(\mathbf{x}, t) &= \phi[\varphi(\mathbf{x}, t), t], \end{aligned} \quad (6)$$

which is a time-dependent field on the reference body, M . We recall from Lecture ~~12~~ 13 that the volume elements in the reference body and physical space are related by $d^3 \mathbf{y} = J(\mathbf{x}, t) d^3 \mathbf{x}$ ~~$d^3 \mathbf{y} = J(\mathbf{x}, t) d^3 \mathbf{x}$~~ with $J = \det(\mathbf{F})$ the Jacobian of the motion. We can then change variables within eq. ~~(4)~~ (4) and use the definition of ζ in eq. ~~(6)~~ (6) to obtain

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3 \mathbf{x}',$$

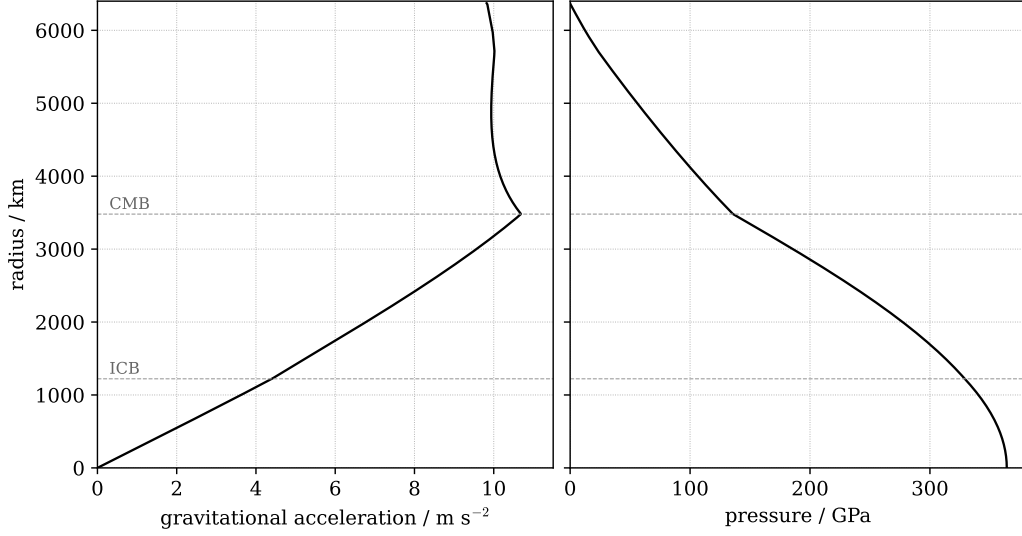


Fig. 1: Gravitational acceleration (left) and pressure (right) as functions of radius in the earth model PREM, calculated from its density profile. The dashed lines mark the core–mantle boundary (CMB) and inner core boundary (ICB).

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3\mathbf{x}', \quad (7)$$

where from Lecture 12–13 we know that conservation of mass requires

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t], \quad (8)$$

with ρ the **referential density**. In a similar manner, we define the **referential gravitational acceleration** by

$$\gamma_i(\mathbf{x}, t) = g_i[\varphi(\mathbf{x}, t), t], \quad (9)$$

which from eq. (5) can be written

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}') [\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d^3\mathbf{x}', \quad (10)$$

Gravitational binding energy

At each instant of time we can associate a **gravitational binding energy** with the elastic body. This is the energy that would be required to assemble the body from matter dispersed at infinity, assuming that its constituent parts are acted on only by their

mutual gravitational attraction. Because gravity is a conservative force, the binding energy is independent of the details of the assembly.

Suppose that the mass of the body is already in place, and that its spatial density and gravitational potential are denoted by ρ and ϕ . To determine an expression for the binding energy we will consider how this quantity changes due to an infinitesimal addition of mass to the body. By definition of the gravitational potential, the work required to bring in an additional mass element $\delta\rho d^3\mathbf{y}$ from infinity to a point $\mathbf{y}$ within the body is $\delta\rho \phi d^3\mathbf{y}$. The ~~total change in the~~ binding energy associated with a perturbation $\delta\rho$ to ~~the~~ body's density ~~by~~ is, therefore, given by

$$\begin{aligned} \delta\mathcal{V}_g &= \int_{M_t} \delta\rho \phi d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= \int_{M_t} \delta\rho \phi d^3\mathbf{y}, \end{aligned} \quad (11)$$

to first-order accuracy. Using the perturbed Poisson's equation $\nabla^2\delta\phi = 4\pi G\delta\rho$ for the resulting change in gravitational potential, we can write this increment to the binding energy as

$$\begin{aligned} \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \\ \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \end{aligned} \quad (12)$$

where the final equality follows from the fact that $\nabla^2\delta\phi = 0$ outside of the instantaneous body M_t . Using the identity

$$\begin{aligned} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial\delta\phi}{\partial x_i} \right) - \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} \\ \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial\delta\phi}{\partial x_i} \right) - \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i}, \end{aligned} \quad (13)$$

and applying the divergence theorem we obtain

$$\begin{aligned} \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} d^3\mathbf{y}, \end{aligned} \quad (14)$$

where we have used the continuity of the potentials and their normal derivatives across ∂M_t , while the surface integral at infinity vanishes because $\phi \sim r^{-1}$ and $\|\nabla\delta\phi\| \sim r^{-2}$ as $r \rightarrow \infty$. It follows that the increment to the gravitational binding energy can be written

$$\delta\mathcal{V}_g = \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\phi}{\partial x_i} d^3\mathbf{y} \right)$$

$$\delta \mathcal{V}_g = \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3 \mathbf{y} \right), \quad (15)$$

and so we arrive at the result

$$\mathcal{V}_g = -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3 \mathbf{y}. \quad (16)$$

Using ~~the Poisson~~ Poisson's equation for ϕ , we can write the integrand of the above expression in the form

$$\frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - \phi \left(\frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - 4\pi G \varrho \phi, \quad (17)$$

and applying the divergence theorem we obtain

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \varrho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3 \mathbf{y}, \quad (18)$$

where, as above, we have used the continuity of ϕ and its normal derivative across ∂M_t , and the surface integral at infinity again vanishes. To include this binding energy into Hamilton's principle, it will be necessary to transform eq. ~~(18)~~ (18) to the following integral over the reference body

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d^3 \mathbf{x}, \quad (19)$$

where we have used the definition of the referential gravitational potential in eq. ~~(6)~~ (6) and the statement of conservation of mass during the deformation in eq. ~~(8)~~ (8).

The equations of motion relative to inertial space

Having obtained an appropriate expression for the gravitational binding energy, the equations of motion can be readily obtained using Hamilton's principle. The Lagrangian for the problem is

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2} \rho(\mathbf{x}) \zeta(\mathbf{x}, t),$$

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t), \quad (20)$$

where we note that the elastic strain energy is allowed to be time-dependent to accommodate a seismic source. An important point about this Lagrangian is that it is **non-local** due to the gravitational binding energy. This means that the value of the Lagrangian at a point and time cannot be expressed only in terms of the values of the various fields and their derivatives at this point. The physical reason for this non-locality is, of course, the **action-at-a-distance** property of Newtonian gravitation.

The action for the body takes the usual form

$$\begin{aligned} \mathcal{S}(\varphi) &= \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d\mathbf{x} dt, \\ \mathcal{S}(\varphi) &= \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x} dt, \end{aligned} \quad (21)$$

and Hamilton's principle requires that this quantity be stationary about the true motion for all perturbations with fixed end points. In obtaining the Euler-Lagrange equations, however, we have to account for the non-local nature of the Lagrangian. As ever, the mathematical statement of Hamilton's principle is ~~that the~~ vanishing of the functional derivative of the action

$$\begin{aligned} \langle DS(\varphi) | \delta\varphi \rangle &= 0, \\ \langle DS(\varphi) | \delta\varphi \rangle &= 0, \end{aligned} \quad (22)$$

for all variations, $\delta\varphi$, subject to the usual fixed-endpoint conditions. Here the ~~the left hand~~ side is, by definition, just the linear term in the expansion of $\mathcal{S}(\varphi + \delta\varphi)$ about φ . Looking at the form of the action, it is clear that this linear term takes the form

$$\begin{aligned} \langle DS(\varphi) | \delta\varphi \rangle &= \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d\mathbf{x} dt. \\ \langle DS(\varphi) | \delta\varphi \rangle &= \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d^3\mathbf{x} dt, \end{aligned} \quad (23)$$

where we have introduced the convenient notation $\frac{\delta\mathcal{L}}{\delta\varphi_i}$, $\frac{\delta\mathcal{L}}{\delta v_i}$ and $\frac{\delta\mathcal{L}}{\delta F_{ij}}$ for the factors multiplying the various perturbations. In the case of a local Lagrangian these factors reduce to normal partial derivatives, and hence it is only the term $\frac{\delta\mathcal{L}}{\delta\varphi_i}$ that requires additional thought. Applying the usual integration by parts steps, we arrive at the ~~Euler-Lagrange equation~~

$$\frac{\delta\mathcal{L}}{\delta\varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta\mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta\mathcal{L}}{\delta F_{ij}} \right) = 0,$$

Euler-Lagrange equation

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta \mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta \mathcal{L}}{\delta F_{ij}} \right) = 0, \quad (24)$$

along with the natural boundary condition

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0$$

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0 \quad (25)$$

on ∂M .

As in the non-gravitating case we have

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}$$

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}, \quad (26)$$

while in the ~~final~~second problem set you will show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i$$

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i, \quad (27)$$

with γ_i the referential gravitational acceleration as defined above. The equations of motion, therefore, take the expected form

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0,$$

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0, \quad (28)$$

with $T_{ij} = \frac{\partial W}{\partial F_{ij}}$ the first ~~Piola Kirchhoff~~Piola-Kirchhoff stress tensor. The form of these equations is not, of course, unexpected. To the balance of linear momentum obtained previously, we have simply added the gravitational force acting on the body due to its own mass.

Use of a co-rotating reference frame

We have obtained the equations of motion for a self-gravitating body with respect to an **inertial reference** frame defined by the “fixed stars”, and these results could be applied directly to the Earth. As, however, the Earth undergoes a nearly steady rotation about its polar axis, it is useful to reformulate the equations of motion with respect to a **non-inertial frame that rotates with the Earth**.

To do this we express the motion as

$$\begin{aligned} \varphi_i(\mathbf{x}, t) &= \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t), \\ \varphi_i(\mathbf{x}, t) &= \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t), \end{aligned} \quad (29)$$

where $\mathbf{R}(t)$ is a rotation matrix. Here ~~$\Phi(t)$ denotes~~ $\Phi(t)$ denotes the translation of the reference frame relative to inertial space, and $\mathbf{R}(t)$ changes of orientation. The term $\bar{\varphi}(\mathbf{x}, t)$ describes the motion relative to the non-inertial frame, and will be called the **relative motion**.

Differentiating eq. (29) with respect to time we see that the velocity relative to inertial space can be written

$$\begin{aligned} v_i &= \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kj} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right) \\ v_i &= \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kj} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right), \end{aligned} \quad (30)$$

where we have used the identity $R_{kj}R_{kl} = \delta_{jl}$ for rotation matrices. In fact, if we differentiate this latter identity we find

$$\begin{aligned} \frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} &= 0. \\ \frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} &= 0, \end{aligned} \quad (31)$$

and hence conclude that $R_{kj} \frac{dR_{kl}}{dt}$ is an **anti-symmetric matrix**. It follows that we can use the alternating tensor to write

$$\begin{aligned} R_{kj} \frac{dR_{kl}}{dt} &= \epsilon_{jkl} \Omega_k \\ R_{kj} \frac{dR_{kl}}{dt} &= \epsilon_{jml} \Omega_m, \end{aligned} \quad (32)$$

for a uniquely determined vector Ω . Using this result, the velocity becomes

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).$$

$$\underline{v_i} = \frac{d\Phi_i}{dt} + R_{ij}(\bar{v}_j + \epsilon_{jkl}\Omega_k\bar{\varphi}_l). \quad (33)$$

Here $\bar{\mathbf{v}}$ is the velocity as seen from the non-inertial frame, while $\underline{\Omega}$ is the **angular velocity of the frame**. Differentiating again and applying the same ideas, the acceleration relative to inertial space can be written

$$\begin{aligned} \frac{\partial v_i}{\partial t} &= \frac{d^2\Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl}\Omega_k\bar{v}_l + \epsilon_{jkl}\frac{d\Omega_k}{dt}\bar{\varphi}_l + \epsilon_{jkl}\epsilon_{lmn}\Omega_k\Omega_m\bar{\varphi}_n \right) \\ \underline{\frac{\partial v_i}{\partial t}} &= \underline{\frac{d^2\Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl}\Omega_k\bar{v}_l + \epsilon_{jkl}\frac{d\Omega_k}{dt}\bar{\varphi}_l + \epsilon_{jkl}\epsilon_{lmn}\Omega_k\Omega_m\bar{\varphi}_n \right)}. \end{aligned} \quad (34)$$

On the right hand right-hand side we can identify various terms. The first is the acceleration of the frame relative to inertial space. Within the brackets we then see, relative to the non-inertial frame, accelerations associated with the motion along with the expected **Coriolis**, **Euler**, and **centrifugal** contributions.

We have not yet specified *how* the non-inertial frame is to be chosen. Within seismology the most common approach is to take

$$\begin{aligned} \underline{\frac{d\Phi}{dt}} &= \underline{0}, \quad \underline{\frac{d\Omega}{dt}} = \underline{0}, \\ \underline{\frac{d\Phi}{dt}} &= \underline{0}, \quad \underline{\frac{d\Omega}{dt}} = \underline{0}, \end{aligned} \quad (35)$$

which corresponds to a frame whose **origin is static** relative to the “fixed stars” “fixed stars” but **rotates steadily**, with the angular velocity chosen so that the frame **approximately co-rotates with the Earth**. The advantage of this choice is its simplicity. But it does mean that orbital, rotational, and relative motion of the Earth are mixed up within the term $\bar{\varphi}$. A preferable approach is to use the so-called **Tisserand frame** which is defined such that the relative motion, $\bar{\varphi}$ carries no net linear **nor angular momentum**¹. or angular momentum¹. But we will stick to the simpler and more common choice in this course.

To complete the derivation of the equations of motion in the non-inertial frame, we note from eq. (29) that

$$\underline{F_{ij}} = \underline{R_{ik}\bar{F}_{kj}}, \quad (36)$$

with $\bar{\mathbf{F}}$ the deformation gradient associated with relative motion. Due to material frame indifference, we then have $W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \bar{\mathbf{F}})$ as should be physically expected. Using the chain rule, we find

$$\underline{\bar{T}_{ij}} = \underline{\frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}})} = \underline{\frac{\partial}{\partial \bar{F}_{ij}}W(\mathbf{R}\bar{\mathbf{F}})} = \underline{\frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}})\frac{\partial}{\partial \bar{F}_{ij}}(R_{km}\bar{F}_{ml})} = \underline{R_{ki}T_{kj}}$$

¹ See “On the elastodynamics of rotating planets” by Maitra & Al-Attar (2024) for details.

¹ See Maitra & Al-Attar (2024) for details.

$$\bar{T}_{ij} = \frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}}) = \frac{\partial}{\partial \bar{F}_{ij}} W(\mathbf{R}\bar{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}}) \frac{\partial}{\partial \bar{F}_{ij}} (R_{km} \bar{F}_{ml}) = R_{ki} T_{kj}, \quad (37)$$

with the ~~left hand~~ left-hand side the stress associated with $\bar{\varphi}$. Passing the rotation matrix to the other side, we have then obtained the useful identity

$$\begin{aligned} T_{ij} &= R_{ik} \bar{T}_{kj} \\ T_{ij} &= R_{ik} \bar{T}_{kj}, \end{aligned} \quad (38)$$

between the first ~~Piola-Kirchhoff~~ Piola-Kirchhoff stress relative to the inertial and non-inertial reference frames. Finally, we put eq.(29 (29)) into eq.(10 (10)) to find the relationship

$$\begin{aligned} \gamma_i &= R_{ij} \bar{\gamma}_i \\ \gamma_i &= R_{ij} \bar{\gamma}_j, \end{aligned} \quad (39)$$

between the referential gravitational acceleration in the two frames, with ~~$\bar{\gamma}$~~ $\bar{\gamma}$ defined exactly as in eq.(10 (10)) but with $\bar{\varphi}$ replacing φ . Using these various results we have shown that the equations of motion are equivalent to

$$\begin{aligned} \rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk} \Omega_j \bar{v}_k + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i &= 0. \\ \rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk} \Omega_j \bar{v}_k + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i &= 0. \end{aligned} \quad (40)$$

These equations are identical in form to those in an inertial frame except for the inclusion of the Coriolis and centrifugal accelerations – the Euler acceleration has been removed because we have restricted attention to a steadily rotating frame. From this point on we will consistently work in the approximately co-rotating frame, and hence will simplify notations by dropping overlines on terms linked to the relative motion.

Linearised equations of motion

As ever, our main concern is with the linearised equations of motion about an equilibrium state. Strictly, we should use the term **relative equilibrium**, as the body is now undergoing a ~~steadily~~ steady rotation relative to inertial space. We consider a disturbance to the body caused by a stress glut, and hence take the strain energy to be

$$\begin{aligned} W(\mathbf{x}, t, \mathbf{F}) &= U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij} \\ W(\mathbf{x}, t, \mathbf{F}) &= U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}, \end{aligned} \quad (41)$$

with s a perturbation parameter, and $\bar{\mathbf{S}}$ a given stress glut. As in Lecture 15, the elastic tensor that appears below is the second derivative of the time-independent part

$U(\mathbf{x}, \mathbf{C})$ of this strain energy function. In a now standard manner we can obtain linearised equations of motion for the displacement vector defined as the first-order term in the expansion

$$\begin{aligned} \varphi_i(\mathbf{x}, t) &= x_i + s u_i(\mathbf{x}, t) + O(s^2), \\ \varphi_i(\mathbf{x}, t) &= x_i + s u_i(\mathbf{x}, t) + O(s^2). \end{aligned} \quad (42)$$

At ~~zeroth-order~~ zeroth order, the equations of motion take the form

$$\begin{aligned} \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \end{aligned} \quad (43)$$

where T_{ij}^0 and γ_i^0 denote the equilibrium values of the stress tensor and gravitational acceleration. We will have more to say about these equations in the following lecture, but for the moment note that there must be a balance between the centrifugal ~~and~~ and gravitational forces on the body and ~~a~~ the resulting equilibrium stress; the simplifying assumption of a stress-free equilibrium is no longer an option.

From the first-order term in the expansion we arrive at the linearised equations of motion for the displacement field

$$\begin{aligned} \rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 &= -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \\ \rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 &= -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \end{aligned} \quad (44)$$

where γ_i^1 denotes the first-order perturbation to the referential gravitational ~~potential~~ acceleration. Using a binomial expansion, this term can be written explicitly as

$$\begin{aligned} \gamma_i^1 &= -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d\mathbf{x}', \\ \gamma_i^1 &= -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}', \end{aligned} \quad (45)$$

where it is understood that primed terms are functions of the integration variable. The boundary conditions for the problem take the expected form

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j.$$

$$\underbrace{A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j}_{\text{action-at-a-distance}} = \bar{S}_{ij} \hat{n}_j. \quad (46)$$

The key point is that the inclusion of gravitation and rotation into the problem only requires some additional terms in the equations of motion considered so far. It is notable, however, that these are no longer partial differential equations, but **integro-partial-differential-integro-differential equations**, with both derivatives and integrals of the unknown functions occurring within the equations of motion. Again, this feature derives from the **action-at-a-distance** property of Newtonian gravitation.

The neglect of self-gravitation and rotation

We have obtained the linearised equations of motion within a steadily rotating and self-gravitating body. It is natural to ask if and why it has been reasonable to neglect these physical phenomena up till this point. To do so, we will perform a scaling analysis of the equations.

We assume that the displacement vector field is characterised by a typical amplitude $\|\mathbf{u}\| \sim U$, a length-scale $\|\mathbf{u}\| \sim U$, a length scale L and time-scale T . The following **order-of-magnitude** scaling estimates are then immediate

$$\begin{aligned} \underbrace{\rho \frac{\partial^2 u_i}{\partial t^2}} &\sim \underbrace{\frac{\bar{\rho} U}{T^2}}, \\ \underbrace{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} &\sim \underbrace{\frac{\bar{\rho} \Omega U}{T}}, \\ \underbrace{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} &\sim \underbrace{2 \bar{\rho} \Omega^2 U}, \\ \underbrace{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} &\sim \underbrace{\frac{\bar{\mu} U}{L^2}}, \end{aligned}$$

$$\underbrace{\rho \frac{\partial^2 u_i}{\partial t^2}} \sim \underbrace{\frac{\bar{\rho} U}{T^2}}, \quad (47)$$

$$\underbrace{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} \sim \underbrace{\frac{\bar{\rho} \Omega U}{T}}, \quad (48)$$

$$\underbrace{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} \sim \underbrace{\bar{\rho} \Omega^2 U}, \quad (49)$$

$$\underbrace{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} \sim \underbrace{\frac{\bar{\mu} U}{L^2}}, \quad (50)$$

where $\bar{\rho}$ and $\bar{\mu}$ denote typical values of density and elastic **moduli** within the body, while Ω is the magnitude of the rotation vector. The scaling estimate for the gravitational term within the equations of motion requires more thought. The referential

gravitational potential at equilibrium is given by

$$\zeta_0 = -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d\mathbf{x}'.$$

$$\zeta_0 = -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d^3\mathbf{x}'. \quad (51)$$

Taking the gradient of this expression twice, we obtain

$$\frac{\partial \zeta_0}{\partial x_i \partial x_j} = G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d\mathbf{x}',$$

$$\frac{\partial^2 \zeta_0}{\partial x_i \partial x_j} = G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d^3\mathbf{x}', \quad (52)$$

and using ~~the Poisson~~Poisson's equation $\nabla^2 \zeta_0 = 4\pi G \rho$ along with eq. ~~(45)~~(45) we arrive at the desired scaling estimate

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U$$

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U. \quad (53)$$

Using these results, we can quantify the relative importance of the terms within the force balance. First we have

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2},$$

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2}, \quad (54)$$

where $\bar{v} = \sqrt{\frac{\bar{\mu}}{\bar{\rho}}}$ $\bar{v} = \sqrt{\bar{\mu}/\bar{\rho}}$ represents a typical elastic wave speed in the model. This ratio scales like L^2 where L is the ~~characteristics length-scale~~characteristic length scale of the deformation. We conclude that for deformations occurring over sufficiently small ~~length-scales~~length scales, gravitational forces will be negligible relative to elastic ones. We should then ask whether gravitational forces will ever be of importance within the Earth? To answer this, we note that the largest-scale deformation within the Earth can have L equal to its average radius ~~6371 km~~6371 km. Recalling that $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and taking as typical values $\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ ms}^{-1}$ we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3.$$

$\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ m s}^{-1}$, we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3. \quad (55)$$

For such planetary scale deformations, it follows that gravitational forces are of the same order of magnitude as elastic forces, and so cannot be neglected within the dynamics.

Next, we consider the ratio of the Coriolis and inertial terms, and find

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T$$

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T. \quad (56)$$

It follows that the Coriolis terms will be negligible for motions whose **time-scale** time scale is short relative to the rotation period of the Earth. A similar conclusion is readily shown to hold for the centrifugal term. It is known that seismic deformation is limited to periods shorter than about one hour, and so rotational effects play a relatively small (but not negligible) role in the problem. There are, however, processes at longer periods, e.g. those associated with solid-Earth tides, for which rotation becomes very important.

The conclusions of this scaling analysis can be checked directly by computing synthetic seismograms in a spherically symmetric earth model with and without self-gravitation, as shown in Fig. 2, where attenuation has been neglected so that the long-period waves persist over the six hours shown.² Over the full frequency band of the calculation, which extends to 50 mHz, the two seismograms are indistinguishable for the first few hours, and they remain so when band-pass filtered to periods shorter than a few minutes, in keeping with the estimate that gravitational forces are negligible for deformations on length scales much smaller than the radius of the Earth. As the pass band is moved to longer periods, however, differences appear and steadily grow, and in the lowest band, at periods between eight and seventeen minutes, they are substantial: the oscillations are slightly but systematically slower without gravitation, so that the two waveforms drift apart as the long-period surface waves circle the Earth. Even at the shortest periods small differences eventually become visible in the surface waves that have circled the Earth several times, since a fractional change in wave speed of only 10^{-4} accumulates into a noticeable phase shift over the hundreds of cycles they have then completed. These are precisely the deformations whose length scale is comparable to that of the planet, and we shall see in Lecture 23 that the periods of the corresponding free oscillations of the Earth are altered by self-gravitation.

² The calculations use the program `yspec`, which solves the equations of motion of this lecture in a spherically symmetric earth model by direct numerical integration in the frequency domain (Al-Attar & Woodhouse 2008).

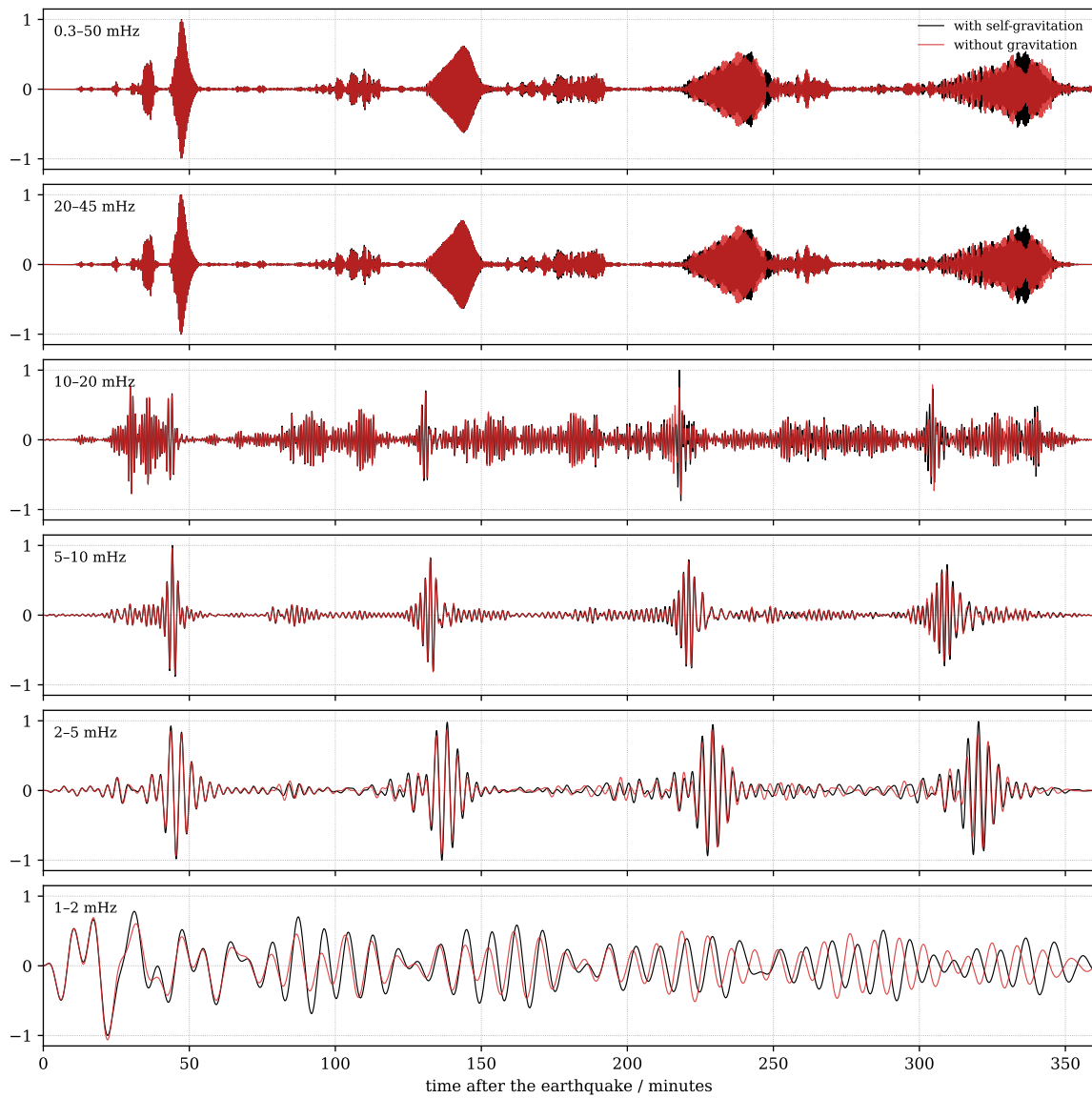


Fig. 2: Synthetic vertical-component acceleration seismograms in the earth model PREM for an earthquake of moment magnitude 8 at 20 km depth, recorded at an epicentral angle of 90°, computed with (black) and without (red) self-gravitation. The top trace shows the full band of the calculation, from 0.3 to 50 mHz, and those below it the same seismogram band-pass filtered in successively lower frequency bands. Each trace is normalised by its maximum. The calculations neglect attenuation, so that the surface waves continue to circle the Earth undiminished over the six hours shown. The two calculations differ only in the lowest bands, and there the differences grow with time as the long-period surface waves circle the Earth.

What you need to know and be able to do

- (i) Understand the relevant parts of Newtonian gravity, including the manner in which the gravitational potential and acceleration can be transformed into quantities defined on the reference body.
- (ii) Know physically what the gravitational binding energy of a planet represents, and how it can be determined.
- (iii) Know how the gravitational binding energy is incorporated into the Lagrangian, and how the equations of motion can be obtained. The full derivation is too long to be examined, but you may be asked to reproduce parts of it given appropriate information and guidance.
- (iv) Perform scaling arguments to determine the relative size of terms ~~with in~~ within an equation, and to determine whether some of them might be negligible. This is a technique required throughout this course, and you should ~~insure~~ ensure you are happy with it.